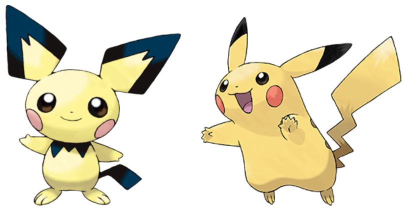
KNN to classify Pikachu and Pichu

The purpose of this exercise is to use the tools you've learned in Python to implement a simplified machine learning algorithm.

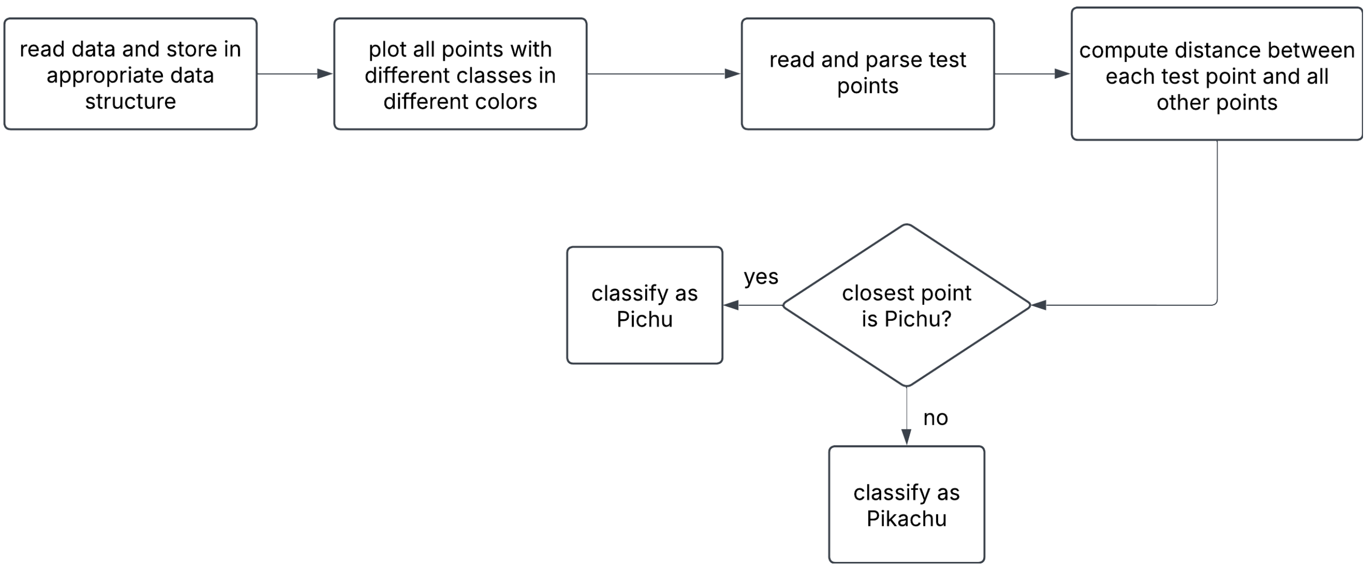


In this lab there is (simulated) data for the heights and widths of Pichu and Pikachu. You will create an algorithm that, based on the given data, can determine whether a new data point should be classified as Pichu or Pikachu.

> Note: You may not use data-processing libraries such as pandas or similar for this assignment. You also may not use machine learning libraries. This lab is intended to give you practice with Python, NumPy, and Matplotlib.

Basic task

Follow this flowchart to build the basic algorithm



Answers for the given test data:

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Sample with (width, height): (25, 35) classified as Pikachu
Sample with (width, height): (15, 14) classified as Pichu
Sample with (width, height): (26, 30) classified as Pichu
Sample with (width, height): (22, 45) classified as Pikachu
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Tasks

Do these tasks after you have correctly classified the test data.

1. Let the user enter a test point and determine its class. Include error handling for negative numbers and non-numeric inputs. Make sure the error messages are user-friendly.

2. The approach we used with the nearest point can misclassify when the points for the respective classes overlap. Now, instead, choose the five nearest points to your test point. The class of the test point is determined by the majority class among the nearest points.

Majority voting

3. Randomly split the original data so that:

   - 90 are training data (45 Pikachu, 45 Pichu)
   - 10 are test data (5 Pikachu, 5 Pichu)

4. Compute the accuracy using the following formula:

$$\text{accuracy} = \frac{\#TP + \#TN}{\text{Total}}$$

where

	Pikachu actual	**Pichu actual**
Pikachu predicted	TP	FP
Pichu predicted	FN	TN

Here we treat Pikachu as positive and Pichu as “non-Pikachu,” i.e., negative.

Also compute recall and precision.